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Education

- 2017 – 2023 **Ph.D., Stanford University** Graduate School of Education
Dissertation title: *Computational Validity*.
- 2019 – 2021 **M.S., Stanford University** Computer Science
- 2010 – 2012 **M.S., Drexel University** Education
- 2001 – 2005 **B.S., Drexel University** Physics

Academic Appointments

- 2023 – **Assistant Professor of Applied Statistics**, New York University,
Steinhardt School of Culture, Education, and Human Development,
Department of Applied Statistics, Social Science, and the Humanities

Publications

Journal Articles

1. Domingue, B. W., **Kanopka, K.**, Ulitzsch, E., & Zhang, L. (2025). Implied probabilities of polytomous response functions for model-based prediction and comparison. *Behaviormetrika*.
2. Domingue, B. W., Rahal, C., Faul, J., Freese, J., **Kanopka, K.**, Rigos, A., Stenhaus, B., & Tripathi, A. S. (2025). The intermodel vigorish (imv) as a flexible and portable approach for quantifying predictive accuracy with binary outcomes. *PloS one*, 20(3), eo316491.
3. Ma, W. A., Richie-Halford, A., Burkhardt, A. K., **Kanopka, K.**, Chou, C., Domingue, B. W., & Yeatman, J. D. (2025). Roar-cat: Rapid online assessment of reading ability with computerized adaptive testing. *Behavior Research Methods*, 57(1), 1–17.
4. Caffarra, S., **Kanopka, K.**, Kruper, J., Richie-Halford, A., Roy, E., Rokem, A., & Yeatman, J. D. (2024). Development of the alpha rhythm is linked to visual white matter pathways and visual detection performance. *Journal of Neuroscience*, 44(6).
5. Domingue, B. W., **Kanopka, K.**, Kapoor, R., Pohl, S., Chalmers, R. P., Rahal, C., & Rhemtulla, M. (2024). The intermodel vigorish as a lens for understanding (and quantifying) the value of item response models for dichotomously coded items. *Psychometrika*, 1–21.
6. **Kanopka, K.**, Claro, S., Loeb, S., West, M., & Fricke, H. (2024). Are changes in reported social-emotional skills just noise? the predictive power of longitudinal differences in self-reports. *AERA Open*, 10.
7. **Kanopka, K.**, & Domingue, B. W. (2024). A position-sensitive mixture item response model. *Journal of Educational and Behavioral Statistics*, 10769986241289399.
8. Kapoor, R., Fahle, E., **Kanopka, K.**, Klinowski, D., Ribeiro, A. C. T., & Domingue, B. W. (2024). Differences in time usage as a competing hypothesis for observed group differences in accuracy with an application to observed gender differences in pisa data. *Journal of Educational Measurement*.
9. Moeller, K. J., **Kanopka, K.**, French, J., Hook, T., & Sedighi, M. (2024). Educational capitalisation: A co-formational feminist framework for conceptualising investment in for-profit education within the racialised and gendered political economy. *Globalisation, Societies and Education*, 1–15.

[†] denotes student collaborators

10. Trejo, S., & **Kanopka, K.** (2024). Using the phenotype differences model to identify genetic effects in samples of partially genotyped sibling pairs. *Proceedings of the National Academy of Sciences*, 121(49), e2405725121.
11. Biernacki, P. J., Altavilla, J., **Kanopka, K.**, Hsieh, H., & Solano-Flores, G. (2023). Long-term english learners' mathematics course trajectories: Downstream consequences of early remediation on college preparation. *International Multilingual Research Journal*, 17(2), 122–138.
12. Ramamurthy, M., Richie-Halford, A., **Kanopka, K.**, Hartsough, A., Osuna, K., Gorno-Tempini, M. L., & Yeatman, J. D. (2023). Can a gamified, rapid, online assessment of letter encoding ability in kindergarten and first grade children predict future reading development? *Journal of Vision*, 23(9), 5858–5858.
13. Ulitzsch, E., Domingue, B. W., Kapoor, R., **Kanopka, K.**, & Rios, J. A. (2023). A probabilistic filtering approach to non-effortful responding. *Educational Measurement: Issues and Practice*.
14. Domingue, B. W., **Kanopka, K.**, Mallard, T. T., Trejo, S., & Tucker-Drob, E. M. (2022). Modeling interaction and dispersion effects in the analysis of gene-by-environment interaction. *Behavior genetics*, 52(1), 56–64.
15. Domingue, B. W., **Kanopka, K.**, Stenhaus, B., Sulik, M. J., Beverly, T., Brinkhuis, M., Circi, R., Faul, J., Liao, D., McCandliss, B., et al. (2022). Speed–accuracy trade-off? not so fast: Marginal changes in speed have inconsistent relationships with accuracy in real-world settings. *Journal of Educational and Behavioral Statistics*, 47(5), 576–602.
16. Domingue, B. W., **Kanopka, K.**, Trejo, S., Rhemtulla, M., & Tucker-Drob, E. M. (2022). Ubiquitous bias and false discovery due to model misspecification in analysis of statistical interactions: The role of the outcome's distribution and metric properties. *Psychological Methods*.
17. Domingue, B. W., **Kanopka, K.**, Stenhaus, B., Soland, J., Kuhfeld, M., Wise, S., & Piech, C. (2021). Variation in respondent speed and its implications: Evidence from an adaptive testing scenario. *Journal of Educational Measurement*, 58(3), 335–363.
18. Yeatman, J. D., Tang, K. A., Donnelly, P. M., Yablonski, M., Ramamurthy, M., Karipidis, I. I., Caffarra, S., Takada, M. E., **Kanopka, K.**, Ben-Shachar, M., et al. (2021). Rapid online assessment of reading ability. *Scientific reports*, 11(1), 6396.

Book Chapters

1. Ruiz-Primo, M. A., **Kanopka, K.**, & Hernandez, P. A. (2025). Measuring, assessing, or just testing student achievement? evidence from some states' testing programs in the united states. In L. Cohen-Vogel, P. Youngs, & J. Scott (Eds.), *Handbook of education policy research* (2nd, pp. 511–542). American Educational Research Association.

Conference Proceedings

1. Mongkhonvanit, K., **Kanopka, K.**, & Lang, D. (2019). Deep knowledge tracing and engagement with moocs. *Proceedings of the 9th international conference on learning analytics & knowledge*, 340–342.

Preprints

1. Ma, W. A., Liu, Y., **Kanopka, K.**, Ma, W., & Domingue, B. W. (2025). A comparison of the predictive performance of continuous and class-based latent trait models. *PsyArXiv*.
https://doi.org/10.31234/osf.io/acwdz_v1
2. Ramamurthy, M., **Kanopka, K.**, Siebert, J. M., Yan, L., Townley-Flores, C., Zegers, M., Pei, F., Bell, P., Luisa, G. T. M., Yeatman, J. D., & Catts, H. W. (2025). The link between vision and reading: A language-agnostic window into heterogeneity in early reading development. *PsyArXiv*.
https://doi.org/doi.org/10.31234/osf.io/esgzs_v3

3. Zhang, L., Liu, Y., Molenaar, J., Dylan Gilbert, **Kanopka, K.**, & Domingue, B. W. (2025). Realistic simulation of item difficulties. *PsyArXiv*. https://doi.org/10.31234/osf.io/jbhxy_v3
4. Deng, S.[†], Siswandjo, J.[†], & **Kanopka, K.** (2024). Mapping the participation of venture capital in education: An investment network perspective. *SocArXiv*. <https://doi.org/10.31235/osf.io/pekyg>
5. Domingue, B., Braginsky, M., Caffrey-Maffei, L., Gilbert, J. B., **Kanopka, K.**, Kapoor, R., Liu, Y., Nadela, S., Pan, G., Zhang, L., et al. (2024). Solving the problem of data in psychometrics: An introduction to the item response warehouse (irw). <https://doi.org/doi.org/10.31234/osf.io/7bd54>
6. Ramamurthy, M., **Kanopka, K.**, Richie-Halford, A., Domingue, B. W., Pei, F., Bell, P., Yan, L., Hartsough, A., Luisa, G. T. M., & Yeatman, J. D. (2024). Design and validation of a rapid visual processing measure for screening reading difficulties in early childhood. *PsyArXiv*. <https://doi.org/10.31234/osf.io/x6ysc>
7. Ramamurthy, M., **Kanopka, K.**, Siebert, J. M., Yan, L., Townley-Flores, C., Zegers, M., Pei, F., Bell, P., Catts, H. W., Luisa, G. T. M., & Yeatman, J. D. (2024). Visual tasks as language-agnostic early identification measures of reading challenges. *PsyArXiv*. <https://doi.org/10.31234/osf.io/ufwsp>
8. Zhang, L., **Kanopka, K.**, Rahal, C., Ulitzsch, E., Zhang, Z., & Domingue, B. W. (2023). The intermodel vigorish for model comparison in confirmatory factor analysis with binary outcomes. *PsyArXiv*. https://doi.org/10.31234/osf.io/tv9bd_v1

Policy Briefs

1. **Kanopka, K.**, Claro, S., Loeb, S., West, M. R., & Fricke, H. (2020). What do changes in social-emotional learning tell us about changes in academic and behavioral outcomes? *Policy Analysis for California Education*.

Presentations

Invited Talks

1. **Kanopka, K.** A position-sensitive mixture item response model. In: Fordham Psychometrics Seminar at Fordham University. 2024.
2. **Kanopka, K.**, & Moeller, K. Following the capital: Using investment networks to understand venture capital's influence on education. In: Society for Research into Higher Education Digital University Network Event on the Political Economy of EdTech. 2023.
3. **Kanopka, K.** Good luck with the blue book! an irt mixture model for position effects. In: Berkeley Evaluation & Assessment Research (BEAR) Seminar at UC Berkeley. 2021.

Conference Presentations

1. **Kanopka, K.** Prediction quality of a range of polytomous models applied to a large data corpus. In: Annual Meeting of the National Council on Measurement in Education. 2025.
2. **Kanopka, K.**, Deng, S.[†], & Lu, Y.[†]. Item pool maintenance in computer adaptive tests: A network approach. In: International Meeting of the Psychometric Society. 2025.
3. **Kanopka, K.**, van Loon, A., Huang, Y.[†], Shen, R.[†], & Tang, H.[†]. Data augmentation for psychometric applications using a mixed subjects design. In: Pacific Quantitative Psychometrics Conference. 2025.
4. **Kanopka, K.**, Zhang, X.[†], & Harel, D. A markov chain monte carlo procedure with simulated annealing for short form creation. In: International Objective Measurement Workshop. 2025.

5. Lu, Y.[†], Huang, Y.[†], & **Kanopka, K.** Assessing the representativeness of llm-generated item responses using latent class analysis. In: Annual Meeting of the National Council on Measurement in Education. 2025.
6. Shen, R.[†], & **Kanopka, K.** Understanding the evolution of individual response process: An exploratory approach. In: International Meeting of the Psychometric Society. 2025.
7. Domingue, B. W., **Kanopka, K.**, Braginsky, M., Zhang, L., Caffrey-Maffei, L., Kapoor, R., Liu, Y., Zhang, S., & Frank, M. The item response warehouse (irw): A large open repository of response data. In: Annual Meeting of the National Council on Measurement in Education. 2024.
8. Domingue, B. W., **Kanopka, K.**, Caffrey-Maffei, L., Kapoor, R., Zhang, S., & Liu, Y. The item response warehouse (irw): A large open repository of response data. In: Annual Meeting of the National Council on Measurement in Education. 2024.
9. Domingue, B. W., **Kanopka, K.**, Ulitzsch, E., & Zhang, L. Implied probabilities of polytomous response functions for model-based prediction and comparison. In: Annual Meeting of the National Council on Measurement in Education. 2024.
10. **Kanopka, K.**, & Harel, D. A lasso approach for short form item selection. In: International Meeting of the Psychometric Society. 2024.
11. Domingue, B. W., **Kanopka, K.**, Kapoor, R., Pohl, S., Chalmers, R. P., Rahal, C., & Rhemtulla, M. Applying the intermodel vigorish to quantify the value of item response modeling. In: Annual Meeting of the National Council on Measurement in Education. 2023.
12. **Kanopka, K.**, & Domingue, B. W. Projecting the performance of polytomous item response models onto a common scale with the intermodel vigorish. In: International Conference on Computational and Methodological Statistics. 2023.
13. **Kanopka, K.**, Domingue, B. W., Ulitzsch, E., Kapoor, R., Pohl, S., Chalmer, R. P., Rahal, C., & Rhemtulla, M. Bookmaking for item responses: Prediction, profits, and the imv. In: International Objective Measurement Workshop. 2023.
14. **Kanopka, K.**, & Hook, T. J. Following the capital: Using investment networks to understand venture capital's influence on education. In: Annual Meeting of the Comparative and International Education Society. 2023.
15. **Kanopka, K.**, Lang, D., & Alvero, A. Weighing algorithmic tradeoffs: Observations from 60,000 admission essays. In: Annual Meeting of the National Council on Measurement in Education. 2023.
16. **Kanopka, K.**, Yeatman, J., & Burkhardt, A. Incorporating response time using drift diffusion models in an online reading assessment. In: Annual Meeting of the National Council on Measurement in Education. 2023.
17. Kapoor, R., **Kanopka, K.**, & Domingue, B. W. Revisiting angrist et al. (2021): Merging international assessment data is not easy. In: Annual Meeting of the National Council on Measurement in Education. 2023.
18. Kapoor, R., Ruiz-Primo, M. A., Hernandez, P., & **Kanopka, K.** Online response process procedures: Affordances and constraints to collect, analyze, and interpret validity information. In: Annual Meeting of the American Educational Research Association. 2023.
19. Ulitzsch, E., Rios, J. A., Kapoor, R., **Kanopka, K.**, & Domingue, B. W. A probabilistic filtering approach to accounting for noneffortful responding. In: Annual Meeting of the National Council on Measurement in Education. 2023.
20. Wang, N., Li, M., **Kanopka, K.**, Dong, D., Hernandez, P., & Ruiz-Primo, M. A. Modeling context characteristics for contextualized assessment: A bayesian contextualized item response model. In: Annual Meeting of the National Council on Measurement in Education. 2023.

21. **Kanopka, K.** Profiling changes in student writing: From failure to success. In: Annual Meeting of the National Council on Measurement in Education. 2022.
22. **Kanopka, K.,** & Domingue, B. W. Good luck with the blue book: Computational approaches to fairness and validity in admissions testing. In: Annual Meeting of the American Educational Research Association. 2022.
23. Moeller, K., & **Kanopka, K.** Who profits from edtech? an intersectional feminist analysis of venture capital investment in education. In: Annual Meeting of the American Educational Research Association. 2022.
24. **Kanopka, K.,** & Domingue, B. Addressing the blue book problem: Application of a position-sensitive irt mixture model to enem 2014. In: International Meeting of the Psychometric Society. 2021.
25. **Kanopka, K.,** & Domingue, B. W. An irt mixture model for item position effects. In: Annual Meeting of the National Council on Measurement in Education. 2021.
26. **Kanopka, K.,** Hernandez, P., Wang, N., Ruiz-Primo, M. A., Li, M., & Minstrell, J. An application of multiple correspondence analysis to the experimental study of item context. In: Annual Meeting of the American Educational Research Association. 2021.
27. Moeller, K., **Kanopka, K.,** & French, J. Venture capitalists as educational actors: Understanding the racialized political economy of silicon valley investments in education technology. In: Annual Meeting of the American Educational Research Association. 2021.
28. Moeller, K., **Kanopka, K.,** & French, J. Venture capitalists as educational actors: Understanding the racialized political economy of silicon valley investments in education technology. In: Annual Meeting of the Comparative and International Education Society. 2021.
29. Ruiz-Primo, M. A., Li, M., **Kanopka, K.,** Hernandez, P., Dong, D., Wang, N., & Minstrell, J. Analysis and review of context in science: Effects in student performance. In: Annual Meeting of the American Educational Research Association. 2021.
30. Alvero, A., Sedlacek, Q., & **Kanopka, K.** A call for critical and constructive data science in teacher education. In: AERA Conference on Educational Data Science. 2020.
31. Claro, S., **Kanopka, K.,** West, M., Loeb, S., & Fricke, H. Exploring the relationship between changes in social-emotional skills and achievement. In: Annual Meeting of the American Educational Research Association (Conference Cancelled). 2020.
32. Ruiz-Primo, M. A., Li, M., Minstrell, J., Dong, D., **Kanopka, K.,** & Hernandez, P. Mapping the characteristics of contexts in science items: The case of forces and motion items. In: Annual Meeting of the American Educational Research Association (Conference Cancelled). 2020.
33. Hernandez, P., Ruiz-Primo, M. A., Zhai, X., & **Kanopka, K.** Validity study of linked items to determine student fundamental ideas. In: Annual Meeting of the American Educational Research Association. 2019.
34. **Kanopka, K.** Addressing defensive objections: Adversarial examples in automatic essay scoring. In: Annual Meeting of the American Educational Research Association. 2019.
35. **Kanopka, K.** Diagnosing non-parallelism in hierarchically contextualized physics assessments. In: Annual Meeting of the National Association for Research in Science Teaching. 2019.
36. Munoz-Najar Galvez, S., **Kanopka, K.,** & Alvero, A. Identifying (dis)continuities in edtech's discourse of invention. In: Text Analysis Across Domains. 2019.
37. Ruiz-Primo, M. A., Li, M., Minstrell, J., Zhai, X., Dong, D., **Kanopka, K.,** & Hernandez, P. Testing the generalization to the domain inference: The use of contextualized clusters of items. In: Annual Meeting of the National Council on Measurement in Education. 2019.

38. Ruiz-Primo, M. A., Li, M., Minstrell, J., Zhai, X., **Kanopka, K.**, Hernandez, P., & Dong, D. Contextualized science assessments: Addressing the use of information and generalization of inferences of students' performance. In: Annual Meeting of the American Educational Research Association. 2019.
39. Zhai, X., Ruiz-Primo, M. A., Li, M., Dong, D., **Kanopka, K.**, Hernandez, P., & Minstrell, J. Using many-facet rasch model to examine student performance on contextualized science assessment. In: Annual Meeting of the American Educational Research Association. 2019.

Awards and Fellowships

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| 2024 | W. Gabriel Carras Research Award , New York University Steinhardt School of Culture, Education, and Human Development
Brenda H. Loyd Outstanding Dissertation Award , National Council on Measurement in Education |
| 2021 | Gerald J Lieberman Fellowship , Stanford University
Summer Internship , Educational Testing Service |
| 2017 | Stanford Graduate Fellowship in Science & Engineering , Stanford University |

Courses Taught

New York University

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| 2024 – | APSTA-GE 2094 (APSY-GE 2524) : Modern Approaches in Measurement
APSTA-GE 2044 : Generalized Linear Models and Extensions |
| 2023 – | APSTA-GE 2352 : Practicum in Applied Statistics: Statistical Computing |

Stanford University

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| 2023 | EDUC 423B (SOC 302B) : Introduction to Education Data Science: Data Analysis |
| 2019 | EDUC 200A : Introduction to Data Analysis and Interpretation |

Professional Activities

Professional Memberships

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| 2017 – | American Educational Research Association |
| 2018 – | National Council on Measurement in Education |
| 2019 – | Psychometric Society |
| 2023 – | Comparative and International Education Society |

Service to Field

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| Editorial | New Directions for Evaluation (Editorial Board) |
| Reviewer | AERA Open
British Journal of Mathematical and Statistical Psychology
Educational Assessment
Educational Evaluation and Policy Analysis
Educational Measurement: Issues and Practice |

Professional Activities (continued)

Educational Researcher

Journal of Leisure Research

New Directions for Evaluation

Proceedings of the National Academy of Sciences

Organized Sessions

2023 **CMStatistics:** Applied statistical and psychometrics issues in measurement (with Daphna Harel)